Lesson 12.1 - The Chi Square Test

> **Video:** Chi Square Test by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The goodness-of-fit test

Does the **observed** distribution of counts across categories match an **expected** distribution?

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- O = observed count, E = expected count, for each category
- **Degrees of freedom** = $k-1$, where k = number of categories
- It is always a **right-tailed** test: a large χ^2 means the observed counts are far from the expected ones

The chi-square distribution is **not symmetric**; it is skewed to the right. The critical value χ^2_α cuts off area α in the right tail:

- calculated χ^2 > critical value → **rejection region** → reject H_0
- calculated χ^2 ≤ critical value → **do-not-reject region**

Example - which weekday has the most absences?

A principal expects absences to be spread **equally** over the 5-day school week. 100 teachers were asked which day had the most absences. Test at $\alpha = 0.05$.

Day	Mon	Tue	Wed	Thu	Fri	Total
Observed O	23	16	14	19	28	100
Expected E	20	20	20	20	20	100

Hypotheses:

- H_0: absences happen with **equal frequencies** (a uniform distribution)
- H_a: absences do **not** happen with equal frequencies

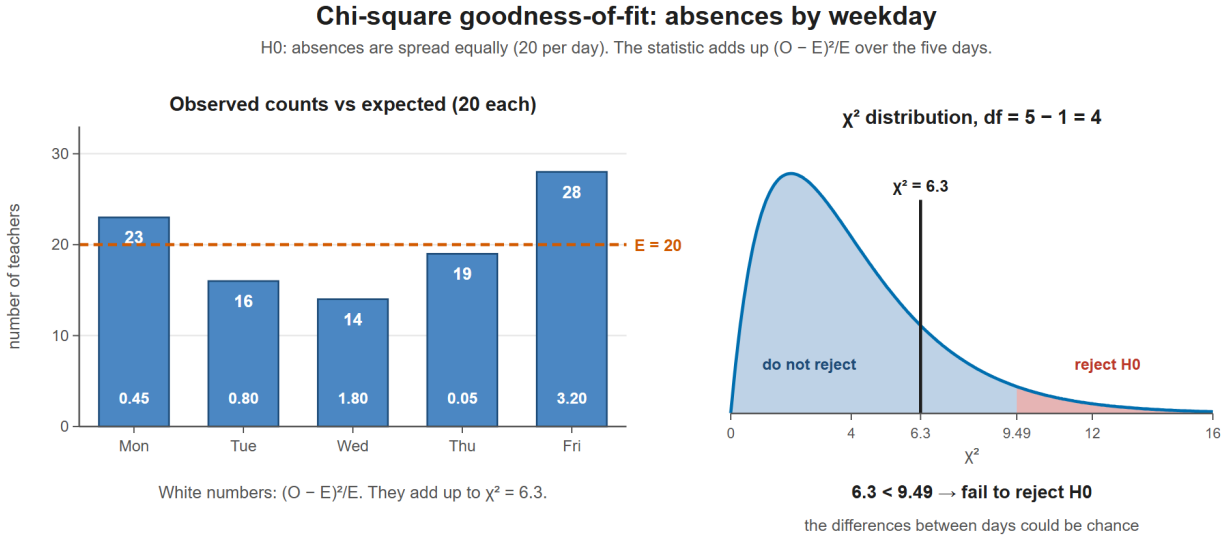
Critical value: df $= 5-1 = 4$. In the χ^2 table with right-tail area 0.05: $\chi^2_{0.05,\,4} = 9.49$.

Test statistic:

Day	$O-E$	$(O-E)^2$	$(O-E)^2/E$
Mon	3	9	0.45
Tue	−4	16	0.80
Wed	−6	36	1.80
Thu	−1	1	0.05
Fri	8	64	3.20
Sum		**126**	**6.30**

$$\chi^2 = \frac{126}{20} = 6.3$$

$6.3 < 9.49$, which is in the do-not-reject region. **Fail to reject H_0.** The variation between days is not large enough to say that absences are unevenly spread across the week.



Left: Friday and Wednesday are furthest from the expected 20, so they contribute the most to χ^2. Right: the total, 6.3, is still below the critical value 9.49, so the uneven pattern could be due to chance.

Key takeaways

- $\chi^2 = \sum(O - E)^2/E$ measures how far the observed counts are from the expected ones.
- Goodness of fit: df = number of categories − 1. It is always right-tailed.
- Large χ^2 (beyond the critical value) → reject H_0 (the data don't fit the expected distribution).